

LACTUL[®] SOLUTION

Lactulose, the active ingredient of Lactul Solution, is a disaccharide consisting of two monosaccharides-galactose and fructose. Lactulose is not only a laxative, it is also used for the treatment of portal-systemic encephalopathy, and it has good tolerance.

Ingredient:

Each ml contains:

Lactulose	670mg
(Sodium benzoate as preservative	0.1%)

Pharmacology (Summary of Pharmacodynamics and Pharmacokinetics):

1. Lactulose is a disaccharide consisting of two monosaccharides-galactose and fructose linked by an oxygen bond. There is no disaccharidase in the brush border of the human intestinal mucosa cells that is capable of splitting lactulose into its constituent monosaccharides. Thus little absorption occurs in the small intestine and lactulose reaches the colon intact.
2. In the ascending colon, lactulose is degraded by the local flora and there is evidence that this process is virtually complete 20-30 minutes after the disaccharide reaches the colon. In the colon, lactulose is broken down to lactic acid (and other short chain carboxylic acid) by the action of colonic bacteria, which exert a local osmotic effect in the colon resulting in the increase in stool volume and frequency observed during lactulose administration.
3. The acid colonic contents convert NH_3 to the ammonium ion $[\text{NH}_4]^+$, trapping it and preventing its absorption. Lactulose causes a decrease in blood ammonia concentration and reduces the degree of portal-systemic encephalopathy.
4. After oral administration, lactulose is poorly absorbed and only small amounts reach the blood. Urinary excretion has been determined to be 3% or less and is essentially complete within 24 hours.

Indication(s):

For the treatment of constipation and hepatic encephalopathy.

Dosage and Administration:

Adults : 15-30ml daily by mouth in a single dose or in 2 divided doses and gradually reduced according to the patient's needs.

Maintenance : 10 to 15ml daily.

Children : 6 - 12 years : 20ml daily.

1 - 5 years : 10ml daily.

< 1 year : 5ml daily

Hepatic encephalopathy:

30-50ml three times daily, subsequently adjusted to produce 2-3 soft stools daily.

In acute cases, a retention enema is given in a proportion of 300ml lactulose to 700ml water, the enema is retained for 30 to 60 minutes, repeated every 4 to 6 hours until the patient is able to take oral medication.

Route of Administration:

Oral

Contraindication(s):

Since lactulose contains galactose (<1.6g/15ml), it is contraindicated in patients who require a low galactose diet.

Precautions / Warning(s) :

1. Lactulose contains galactose and lactose, it should be used with caution in diabetics.
2. Since Lactulose contains lactose (<1.2g/15ml) it should be used with caution in patients with lactose intolerance.
3. Use in pregnancy and lactation:
Reproduction studies in rats and teratologic experiments in both mice and rats did not reveal significant effects on conception, gestation, litter size, or incidence of malformations. However, the use of lactulose during pregnancy requires that the potential benefits be weighed against the potential risks.
4. It is not known whether this drug is excreted in human milk. Caution should be exercised when it is administered to a nursing woman.

Drug Interaction(s):

Concurrent administration of lactulose with antacid may inhibit the desired lactulose - induced drop in colonic pH. Therefore, a possible lack of desired effect of treatment should be taken into consideration before such drugs are given concomitantly with Lactul Solution.

Side Effect(s) / Adverse Reaction(s):

Initial dosing may produce flatulence and intestinal cramps, which are usually transient. Excessive dosage can lead to diarrhea with potential complication such as loss of fluids, hypokalemia, and hypernatremia. Nausea and vomiting have been reported.

Symptoms and Treatment for Overdosage and Antidotes:

There have been no reports of accidental overdosage. In the event of overdosage it is expected that diarrhea and abdominal cramps be the major symptoms. Medication should be terminated.

Storage Condition(s):

Keep in a tight container. Store at temperature below 30°C. Protect from light and moisture.

Shelf-life:

3 years from the date of manufacture.

Product description and Packing(s):

A clear, slightly yellow to amber solution.

Plastic bottles of 60mL, 100mL, 200mL and 500mL.

Plastic bottles of 1L and 4L(for export only).

LARUTAN LACTUL®

Bahan aktif:

Setiap ml mengandungi:

Lactulose	670mg
(Sodium benzoate sebagai bahan pengawet	0.1%)

Indikasi:

Untuk rawatan sembelit dan ensefalopati hepatic.

Dos:

Dewasa : 15-30ml sehari dimakan sebagai dos tunggal atau dalam
pembahagian 2 dos,dan dos dikurangkan secara beransur-ansur mengikut keadaan pesakit.
Dos pengekaln : 10-15ml sehari.

Kanak-kanak: 6-12 tahun : 20ml sehari
1-5 tahun : 10ml sehari
< 1 tahun : 5ml sehari

Ensefalopati hepatic:

30-50ml tiga kali sehari, dos kemudian diselaraskan untuk menghasilkan 2-3 ketul najis yang lembut sehari.

秘通 液

成份:

每ml含:

Lactulose	670mg
(Sodium benzoate防腐劑	0.1%)

適應症:

治療便秘及肝性腦病變。

用法.用量:

成人 : 起始量 : 每天 15 ~ 30ml, 分1至2次服用。
維持量 : 每天 10 ~ 15ml。

兒童 : 6至12歲 : 每天 20ml。
1至5歲 : 每天 10ml。
少以1歲 : 每天 5ml。

肝性腦病變疾患:每天3次, 每次30 ~ 50ml。隨後調整劑量直到每日排出2 ~ 3塊軟便。



Manufacturer and Product Registration Holder:

Y.S.P. INDUSTRIES (M) SDN. BHD. (199001001034)

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